Contents- Unit-IV
Bitcoin

Bitcoin:

- Centralization vs. Decentralization
- Distributed consensus
- Consensus without identity using a block chain
- Incentives and proof of work

Here in this unit we try to explore how Bitcoin achieves decentralization through a mix of technical mechanisms and incentive engineering. Key points include:

1. **Decentralization vs. Centralization**: Bitcoin balances these paradigms, with no single authority controlling the system.
2. **Distributed Consensus**: Nodes agree on transaction validity using a blockchain, ensuring a shared ledger without central control.
3. **Proof-of-Work (Mining)**: Miners compete to solve cryptographic puzzles to add blocks, earning rewards (new bitcoins and fees). This secures the network and prevents Sybil attacks.
4. **Incentives**: Block rewards and transaction fees motivate honest participation.
5. **Security**: A 51% attack (controlling most mining power) could disrupt consensus but cannot steal coins or alter transaction history due to cryptographic safeguards.

Bitcoin's design ensures trustless, decentralized operation by aligning economic incentives with network security.

Centralization vs. Decentralization

- **Decentralization** = No single entity controls everything (e.g., Bitcoin, email).
- **Centralization** = Controlled by one or a few entities (e.g., Facebook, banks).
- Decentralization is an important concept in many technologies, including Bitcoin.
- Key conflict is between centralized systems (controlled by a single entity) and decentralized systems (controlled by many independent participants).

Examples:

1. **Internet** – Mostly decentralized (no single owner).
2. **Email (SMTP)** – Decentralized (anyone can run a server), but big providers (Gmail, Yahoo)
3. **Social Media (Facebook, LinkedIn)** – Centralized (one company controls everything).
4. **Bitcoin** –Decentralized in theory, but some parts (like mining) are more centralized due to high costs.

Most systems are not fully centralized or fully decentralized; they exist on a spectrum.

How Decentralization Relates to Bitcoin

- **Bitcoin is decentralized**, but some parts of the ecosystem may still have centralized elements.
- **Examples of centralization in Bitcoin:**
  - **Bitcoin exchanges** (where you trade Bitcoin for other currencies).
  - **Bitcoin wallets** (software to manage your Bitcoin

Bitcoin's Decentralization Questions:

1. **Who keeps the transaction ledger?** → Everyone (nodes in the network).
2. **Who decides if a transaction is valid?** → Miners & nodes (through consensus).
3. **Who creates new bitcoins?** → Miners (by solving puzzles).
4. **Who changes Bitcoin's rules?** → Developers & community (but changes must be widely accepted).
5. **What gives Bitcoin value?** → People's trust & demand (like gold or stocks).

Decentralization in Different Aspects of Bitcoin

- **Bitcoin Network (Highly Decentralized)**:
  - Anyone can run a Bitcoin node (a computer that supports the Bitcoin system).
  - Thousands of nodes exist worldwide.
- **Bitcoin Mining (Becoming Centralized)**:
  - Anyone can mine Bitcoin, but it requires expensive hardware.
  - This has led to power being concentrated among a few big mining groups.
- **Bitcoin Software Updates (Partially Centralized)**:
  - The Bitcoin software is open-source, but most nodes run the "reference" version.
  - Developers of this version have significant influence over system change

Distributed Consensus

Let’s First see Distributed Consensus vs. Consensus

1. Consensus (General Meaning)

•**Definition:** A group agrees on a single decision.

•**Example:** Friends deciding where to eat.

•**Key Features:**

- Can be **centralized** (e.g., a leader decides).
- Works in **small, trusted groups** (no malicious actors).

2. Distributed Consensus

•**Definition:** Many computers (nodes) agree on data **without a central leader**.

•**Example:** Bitcoin miners agreeing on transaction history.

•**Key Features:**

- **Decentralized** (no single authority).
- Must handle **malicious nodes, delays, failures**.
- Harder because of **Byzantine Generals Problem** (traitors can lie).

What is Distributed Consensus?

- **Goal:** Get all honest computers (nodes) in a network to **agree** on something (e.g., Bitcoin transactions).
- **Problem:** Some nodes might be **malicious** or **fail**, and the network is **slow/unreliable**.

How Bitcoin Uses Consensus:

1.Transactions are Broadcast

1. When Alice pays Bob, she sends the transaction to **all Bitcoin nodes**.
2. Nodes collect these transactions into a **pool** (waiting to be confirmed).

2.Blocks = Batches of Transactions

1. Every ~10 minutes, nodes group transactions into a **block**.
2. Nodes must agree on **which block** gets added to the **blockchain** (the public ledger).

3.Challenges:

1. **No central authority** to decide.
2. **Network delays** mean nodes might see transactions in different orders.
3. **Malicious nodes** could try to cheat (e.g., double-spending).

Why is Distributed Consensus Important?

- Used in computer networks, social media platforms, and blockchain technology.
- Helps keep records synchronized across multiple computers.
- Prevents fraud and ensures a single, agreed-upon version of the data.

How Does Distributed Consensus Work?

- Multiple nodes propose their own version of a record (block).
- A protocol decides which version is valid.
- Honest nodes must agree on a single value (transaction record).
- The chosen value must come from an honest node.

Distributed Consensus in Bitcoin

- Bitcoin is a **peer-to-peer** network where transactions are broadcast to all nodes.
- Nodes must **agree** on:
  - Which transactions are valid.
  - The order in which they happened.
- Transactions are grouped into **blocks**, and consensus happens block by block.
- Some nodes may have different versions of pending transactions due to network delays.

Challenges in Achieving Consensus

- **Network Imperfections:** Internet delays, disconnected nodes.
- **Malicious Nodes:** Some nodes may try to cheat the system.
- **No Global Time:** Different nodes may see events happening in different orders

Consensus Problems

- **Byzantine Generals Problem:** If 1/3 of nodes are dishonest, agreement is impossible.
- **Fischer-Lynch-Paterson Theorem:** With even one faulty node, perfect consensus is impossible in certain conditions.

Bitcoin's Unique Approach to Consensus

- **Incentives:** Bitcoin miners are rewarded for following the rules.
- **Randomization:** Bitcoin's algorithm relies on randomness (Proof of Work).
- **No Fixed Consensus Time:** Transactions become more reliable as time passes.

Consensus without identity using a block chain

What It Means:

- In traditional systems (like banks), you need verified identities to participate.
- In blockchain (like Bitcoin), no real-world identity is needed—just a digital key.
- Nodes (computers) agree on transactions based on rules, not who they are.
- Consensus Without Identity means that participants in the network do not need to reveal or verify their real-world identity to participate in the consensus process.
- Using a Blockchain ensures that all transactions and blocks are verified and agreed upon without needing a central authority or known identities.

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In Bitcoin, anyone can mine or validate transactions without revealing their identity. The network trusts the cryptographic process, not the person behind it.

Consensus without identity using a block chain

How Does It Work?

1. **Anonymous Participation** – Anyone can join and validate transactions without registering or proving their identity.
2. **Proof-Based Verification** – Instead of trusting identities, the system uses cryptographic proofs (e.g., Proof of Work in Bitcoin) to validate transactions.
3. **Decentralized Agreement** – Nodes in the network follow a predefined consensus algorithm to agree on the valid state of the blockchain.
4. **Security Through Economics** – Miners or validators are incentivized to follow the rules because dishonest behavior leads to financial loss.

Simple Analogy:

- **With Identity** = Signing a bank check (needs your name).
- **Without Identity** = Dropping cash into a locked box—nobody knows who put it in, but everyone agrees on the total amount.

Consensus without identity using a block chain

Blockchain achieves agreement (consensus) **without needing real-world identities**, using cryptography and incentives instead

Consensus Without Identity in Bitcoin

•**Bitcoin nodes lack long-term identities** because there is no central authority to assign them.

•**Sybil attack:** A malicious actor can create many fake nodes to appear as multiple participants.

•**Pseudonymity is a goal of Bitcoin:** Users do not need to reveal their real identities to participate.

•**Challenges without identities:**

- Harder to design instructions like "Node with lowest ID should act."
- Security assumptions become difficult without a way to limit the number of malicious nodes.

Solution: Random Node Selection

•System similar to a lottery is assumed to pick a random node fairly.

•System ensures that a malicious actor creating many fake nodes only gets one participation token.

Consensus without identity using a block chain

Implicit Consensus

•Instead of voting, a randomly selected node proposes the next block.

•If a block is valid, other nodes accept it by building on top of it.

•If a block is invalid (malicious), nodes ignore it and build on the previous valid block.

Bitcoin Consensus Algorithm

1.Transactions are broadcast to all nodes and each node collects transactions into a block.

2.A random node is selected to propose a block other nodes accept the block if transactions are valid.

3.Acceptance is shown by referencing the block's hash in new blocks.

Security Against Attacks

1. Stealing Bitcoins

•Impossible since every transaction requires a valid cryptographic signature.
•Without the owner's signature, a hacker cannot spend their coins.

2. Denial of Service Attack

•A malicious node may refuse to include a user's transactions.
•However, the user's transactions will be included when another (honest) node proposes a block.
•This makes the attack ineffective beyond temporary inconvenience.

Consensus without identity using a block chain

3. Double-Spend Attack

•A user (Alice) sends Bitcoin to a merchant (Bob) and broadcasts the transaction.

•An honest node includes this transaction in a block.

•If Alice controls next block, she can create an alternate block that sends same Bitcoin to herself instead.

•The network may then choose the longest valid chain, possibly favoring Alice's version.

This could result in Bob not actually receiving the Bitcoin.

Scenario:

1.Alice pays Bob **1 BTC** for a laptop.

2.At the same time, she tries to **send the same 1 BTC to herself**.

3.If both transactions reach different nodes, who decides which one counts?

Bitcoin's Solution:

•Miners pick **one transaction** per block (first-come, first-served).

•After **6 confirmations** (~1 hour), the transaction is "locked in."

Consensus without identity using a block chain

How Merchants Protect Themselves

- Bob should **wait for multiple confirmations** before accepting payment.
- A transaction in a block that has several blocks built on top of it is much harder to reverse.
- **Zero-confirmation transactions** (where merchants accept payments immediately) are risky

Incentives and proof of work

1.Incentives in Bitcoin

- Bitcoin's decentralization relies on both technical mechanisms and incentive engineering.
- Nodes (participants) need incentives to behave honestly.
- Instead of punishing dishonest nodes (which is hard), Bitcoin **rewards** nodes for creating valid blocks.

2. Block Reward

The node that successfully creates a block gets a **block reward** (newly created bitcoins).

This reward is given through a special **coin-creation transaction** in the block.

Initially, Bitcoin’s block reward was **50 BTC**, then halved to **25 BTC**, and continues halving every **210,000 blocks (~4 years)**.

The total supply is limited to **21 million BTC** (expected to run out by **2140**).

Since only valid blocks are accepted, miners are incentivized to follow the rules.

Bitcoin’s incentives and PoW turn mining into a profitable, fair competition that keeps the system secure

Incentives and proof of work

3. Transaction Fees

As block rewards decrease, miners rely on **transaction fees** as an additional incentive.

Users voluntarily add fees to transactions to ensure faster processing.

Over time, transaction fees will become the primary incentive for miners.

4. Problems in the Consensus Mechanism

Random Node Selection Problem: How to pick a truly random node to create a block?

Mining Rush: The incentive system may cause too many nodes to compete.

Sybil Attack Risk: A malicious user could create many fake nodes to take control.

5. Proof of Work (PoW)

Bitcoin solves the above problems using **Proof of Work (PoW)**.

Instead of selecting a random node, PoW **makes miners compete** using computational power.

The more computing power a miner has, the higher their chance of creating the next block.

This system discourages Sybil attacks since mining requires expensive resources

Bitcoin's incentives and PoW turn mining into a profitable, fair competition that keeps the system secure

Incentives and proof of work

6. Hash Puzzles (Mining Process)

Miners must solve a **hash puzzle** by finding a special number (**nonce**).

The nonce, when combined with the block data and hashed, must produce a result **below a target value**.

The only way to solve this is through **trial and error**.

The difficulty is adjusted to ensure new blocks are created roughly every **10 minutes**.

Once a miner finds a valid nonce, they broadcast the block, and other nodes verify it.

7. Key Properties of Hash Puzzles

Difficult to Compute: Miners must try many different nonces, requiring high computational power.

Decentralized Selection: No central authority decides who mines the next block.

Prevents Spam/Sybil Attacks: Mining requires resources, preventing mass fake nodes.

Bitcoin's incentives and PoW turn mining into a profitable, fair competition that keeps the system secure

Incentives and proof of work

Why Incentives Matter

Problem: Need miners to stay honest without central control.

Solution: Pay them in Bitcoin to follow the rules!

Real-Life Example:

Like paying gold miners for finding gold—they're motivated to work hard and play fair.

Two Ways Miners Earn

1 Block Reward

New Bitcoins created for miners who add valid blocks.

Halving Event: Reward cuts in half every ~4 years (now 3.125 BTC per block).

Total Supply Capped at 21M BTC (last coin mined ~2140).

2 Transaction Fees

Users pay small fees to prioritize transactions.

Miners collect these fees when they add transactions to blocks.

Example:

Like a toll road—you pay a fee to use it, and the toll collector (miner) keeps it.

Bitcoin's incentives and PoW turn mining into a profitable, fair competition that keeps the system secure

Thank You